

Detecting Clickbait with RoBERTa: Improving Classification over Traditional Baselines

Samuel Lee
UC San Diego
hsl023@ucsd.edu

Michael Pena
UC San Diego
mepena@ucsd.edu

Anna Lee
UC San Diego
aml027@ucsd.edu

Sravya Chittuluri
UC San Diego
schittuluri@ucsd.edu

Gillian Wentzel
UC San Diego
gwentzel@ucsd.edu

Abstract

In today’s social media culture, engagement and clicks directly drive revenue, incentivizing creators to use sensationalized or misleading “clickbait” titles. While these headlines initially attract the desired attention towards the post or article, they also erode the reader’s trust and contribute to the spread of misinformation online. Oftentimes, users will come to a judgment based solely on the title without reading the rest of the article and understanding any nuance in the situation. This misrepresentation can then become more well-known than the actual information. Multiple instances of this cycle of getting tricked by clickbait titles eventually lead to a general distrust in journalism. Having the ability to detect clickbait can help improve online content moderation and media literacy. Our aim is to fine-tune a language model to accurately identify whether an article title is clickbait or not. We found a dataset containing article titles from various news outlets, and preprocessed the data using natural language processing techniques like tokenization and removing special characters. Using classical models like logistic regression and random forest as our baselines, we finetuned pretrained LLM transformer models like BERT, DistilBERT, and RoBERTa-base. We achieved more accurate results than the baseline models trained on this task, proving that large language models are a viable solution to identifying clickbait.

1 Introduction

At best, clickbait headlines entice users into reading an article with little substance, increasing the site’s potential virality. At worst, clickbait can be used to scam readers. The Better Business Bureau warns that clicking on bait titles could “allow cyber criminals to hijack your account or steal personal information” (Suciu, 2020). From pushing harmful narratives about cultural events to causing direct

harm to readers, clickbait has become a pressing issue.

In Bazaco et al. (2019), the term “clickbait” is given a qualitative definition with attached variables, which is then used to compare the content of thousands of articles posted on several established newspapers. These variables include incomplete information, expressions, repetition, exaggeration, and soft news. The study found the presence of typical clickbait features in around half the articles, pointing to a dependency on the practice. The article attempts to contextualize this, stating the increasing necessity of internet foot traffic as people become more dependent on social media. To these sites, content manipulation is the best way to maximize attention and guarantee engagement. The presence of clickbait, however, tends to make information less reliable, increasing the desirability of a classifier that can correctly differentiate between clickbait and reality.

Chakraborty et al. (2016) performed linguistic analysis on many news articles to identify features that denoted clickbait titles, and classified headlines by using various Naive Bayes classifiers and support vector machines. They created a browser extension that warns readers of potential clickbait and allows them to block those headlines if they want to. The extension selected which headlines to block based on the linguistic patterns of the articles that readers were interested in reading. Through this, they achieved 93% accuracy in detecting and 89% accuracy in blocking clickbait.

Pothast et al. (2016) populated a corpus of tweets that contained news headlines from various publishers. They then took a different approach by using the teaser message, meta info, and linked web page as features to train logistic regression, random forest, and naive Bayes models. Both of these papers show various methods of classifying clickbait vs. real headlines using classical machine learning. However, we want to see if we can repro-

duce these results using large language models, and potentially achieve the same, if not better results.

Wang et al. (2025) attempted to prompt GPT-3.5 and GPT-4 to predict if a given title is clickbait. They used prompt engineering to provide examples of clickbait vs non-clickbait titles in both English and Chinese. In the end, the LLMs did not perform as well as existing transformer models. Therefore, we want to explore if finetuning large language models will provide better results.

2 Method

2.1 Dataset

We utilized the News Clickbait Dataset from Kaggle, which contains 32,000 news headlines drawn from a variety of online news outlets. Each example consists of a short text string in a headline field and an associated binary label in a clickbait field, indicating whether the headline is judged to be either clickbait (using a 1) or non-clickbait (using a 0). This framing lets us treat the problem as a binary sentence-level classification problem task, where the model sees only the headline text and must decide if it should be labelled as clickbait.

For our experiments, we stored the data in a single CSV file matching this format and used it as the single supervision signal for all models. We focused specifically on the headline text and did not use any additional article content, metadata, or user interaction statistics, so all models should only rely on the lexical and stylistic views present in the titles for this classification.

2.2 Data Preprocessing

All text preprocessing is handled in our project by a shared cleaning function defined in `dataset.py`. Before any modeling, we load the raw CSV file, keep only the headline text and binary label columns, and drop rows with missing or malformed values. The label column is cast to integer form to indicate the assigned clickbait label discussed earlier, so it can be used directly as a target in both scikit-learn and Hugging Face classifiers.

Headlines are then passed through a lightweight normalization step made to preserve clickbait-relevant punctuation, but still removing noisy characters. To specify:

1. Converts all text to lowercase
2. Removes digits, most punctuation, and special characters

3. Preserves exclamation marks and question marks, decidedly kept due to them being common stylistic cues in clickbait

4. Collapses repeated whitespace into single spaces

This processed headline serves as the standardized version used in all subsequent stages of the pipeline. The same preprocessing function is applied to both when we build TF-IDF features for the classical baseline and when we tokenize the text for transformer models, so differences in performance can be attributed to the modeling choices rather than inconsistent input cleaning. We also purposefully avoided heavier linguistic processing, such as lemmatization or stopword removal, since those could delete stylistic markers (pronouns, function words, punctuation patterns, etc.) that are potentially informative for distinguishing clickbait from non-clickbait.

2.3 Baseline Models

2.3.1 Logistic Regression

As a fundamental baseline, we trained a logistic regression classifier using TF-IDF features extracted from the cleaned headlines. Logistic regression models the probability of each class by applying a sigmoid function to a linear combination of input features, making it both computationally efficient and easily interpretable. This baseline provides a clear reference point against which we can compare the performance of our transformer models.

2.3.2 Random Forest

We further included a Random Forest classifier as a non-linear baseline. Like logistic regression, this model is also trained on TF-IDF representations of the cleaned headlines; however, Random Forest consists of an ensemble of decision trees that each learn different sections of the feature space. By combining these different predictions, the Random Forest model is able to capture complex non-linear relationships that a linear model, like Logistic Regression, may overlook. Thus, this model provides a stronger traditional machine learning benchmark for our transformer models.

2.4 Transformer Models

2.4.1 BERT

BERT is a bidirectional transformer encoder pre-trained using masked language modeling and next

sentence prediction (Devlin et al., 2019). Unlike traditional models that process tokens in strictly one direction, BERT analyzes each token in a sequence by simultaneously considering all preceding and following tokens. This bidirectional attention allows the model to capture more subtle semantic cues in the sequence, so fine-tuning BERT on our labeled dataset could lead to better recognition of the specific patterns and linguistic markers that would aid in clickbait classification.

2.4.2 DistilBERT

DistilBERT is a compact version of the original BERT model designed to offer a balance between efficiency and performance (Sanh et al., 2020). Created using a process called knowledge distillation, where a larger model is used to train a smaller model, DistilBERT retains a vast majority of BERT’s performance while requiring significantly less memory and computational power. These characteristics make DistilBERT an attractive option for training and fine-tuning with different hyperparameters.

2.4.3 RoBERTa

RoBERTa is an optimized, more robustly trained variant of BERT. It is trained on larger datasets, with longer input sequences, larger batches, and dynamic masking strategies that change the masked token at each training step (Liu et al., 2019). Furthermore, by removing the next-sentence prediction task and focusing solely on the masked language modeling objective, RoBERTa can have improved understanding and learn richer contextual representations of tokens. Given these enhancements, we expect this model to yield a higher performance on clickbait detection by being able to better understand different language nuances.

2.5 Fine-Tuning Procedure

We used a unified sweep configuration implemented through Weights & Biases to systematically explore and identify the most effective hyperparameters for all our models (Logistic Regression, Random Forest, BERT, DistilBERT, RoBERTa). By keeping the sweep setup consistent across all the models, we ensured that any observed differences in performance were due solely to the models themselves rather than any methodological discrepancies.

To make the search more computationally efficient, especially for the more robust transformer-

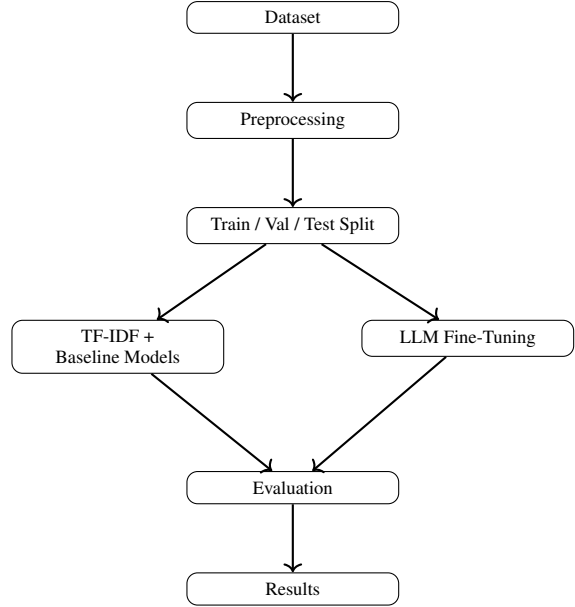


Figure 1: Simplified project workflow for clickbait classification.

based models, all sweeps were conducted on a curated 2,000-sample subset of the training dataset and limited to 3 epochs. This reduced scale allowed us to evaluate a wide range of hyperparameter combinations at a lower cost than training with the entire dataset.

For the transformer models, the sweep explored different values for the batch size, learning rate, maximum sequence length, and warmup ratio (See Table 1). Each combination is trained end-to-end with a binary classification head that seeks to optimize binary cross-entropy loss using the AdamW optimizer.

For the baseline models, the sweep adjusts TF-IDF parameters (maximum document frequency, minimum document frequency, and maximum n-gram) in addition to the classifier-specific settings. For logistic regression tuning, that means exploring various regularization strengths and maximum number of iterations, while for Random Forest, that means exploring different numbers of estimators, maximum tree depth, and minimum number of samples in an internal node (see Table 2 and 3).

After identifying the top-performing hyperparameters for each model based on validation F1, we re-trained the selected configurations on the full training dataset and used 5 epochs to allow for deeper convergence. This two-stage pipeline ensures computational efficiency and a strong final performance for all models.

Hyperparameter	Search Space
Batch Size	{16, 32}
Learning Rate	{1e-5, 2e-5, 3e-5, 5e-5}
Max Sequence Length	{96, 128}
Warmup Ratio	{0, 0.06}
Epochs	3 (fixed)
Model Type	{RoBERTa-base DistilBERT BERT-base}
Rows Used	2000 (fixed subset)
Search Objective	Maximize Eval F1

Table 1: Hyperparameter tuning configuration for transformer models fine-tuning using grid search.

Hyperparameter	Search Space
LR Regularization Strength C	{0.25, 0.5, 1, 2, 4}
LR Max Iterations	{1000, 2000}
TF-IDF Max Document Freq	{0.85, 0.90, 0.95}
TF-IDF Min Document Freq	{1, 2, 5}
TF-IDF Max N-gram	{1, 2}
Rows Used	2000 (fixed)
Objective Metric	Maximize Eval F1

Table 2: Hyperparameter tuning configuration for the logistic regression baseline model.

3 Results

For each model, we evaluated on held-out validation and test splits of the News Clickbait Dataset described earlier. For every model, report accuracy, precision, recall, and F1, and additionally binary cross-entropy loss (when applicable from the training framework). We treat the clickbait label as the positive class, and hyperparameters as well as checkpoints are selected using validation F1, since our goal is to maximize F1 while minimizing loss. Accuracy is reported as a secondary summary metric.

3.1 Logistic Regression Baseline

Our first baseline is a logistic regression classifier trained on TF-IDF features (see Table 4). The validation and test scores were ultimately very similar, with accuracy around 0.89 and F1 around 0.86 on both splits. Precision and recall are both in the mid-0.8 range, indicating that the model both identifies a large population of clickbait headlines and maintains a relatively low rate of false positives. The proximity of validation and test values overall suggests that the model’s performance is stable across unseen data.

Hyperparameter	Search Space
RF Number of Estimators	{200, 400, 800}
RF Max Depth	{None, 20, 40}
RF Min Samples Split	{2, 5}
TF-IDF Max Document Freq	{0.85, 0.90, 0.95}
TF-IDF Min Document Freq	{1, 2, 5}
TF-IDF Max N-gram	{1, 2}
Rows Used	2000 (fixed)
Objective Metric	Maximize Eval F1

Table 3: Hyperparameter tuning configuration for the Random Forest baseline model.

Split	Accuracy	Precision	Recall	F1
Val	0.8870	0.8418	0.8679	0.8546
Test	0.8903	0.8473	0.8699	0.8585

Table 4: Logistic Regression Performance

3.2 Random Forest Baseline

The second baseline is a random forest classifier trained on TF-IDF features (see Table 5). On both splits, the random forest reaches an accuracy in the high-0.87 range and F1 in the low 0.84 range. Precision and recall are relatively balanced, with recall slightly lower than precision, meaning the model is somewhat more likely to miss some clickbait examples than to produce false positives. As with logistic regression, validation, and test scores are close, indicating consistent behavior on held-out data as well.

3.3 RoBERTa Transformer Model

As mentioned before, throughout testing, we had considered three pretrained transformer encoders from the BERT family: BERT-base-uncased, DistilBERT-base-uncased, and RoBERTa-base. Each model was fine-tuned on the clickbait classification task using the same training/validation splits and optimization setup. Across these runs, one setup of RoBERTa-base consistently achieved a very slightly higher F1 score compared to other BERT/DistilBERT setups, with the former having a validation F1 of 0.95532 and the latter at 0.95501, and slightly higher accuracy on both the validation and test sets. Therefore, we selected the RoBERTa-base transformer as our main model, fine-tuned for binary classification. Table 6 reports its validation and test metrics.

On the validation set, RoBERTa reaches an accuracy of 0.9053 and an F1 of 0.8720, with high precision (0.9033) and slightly lower recall (0.8428).

Split	Accuracy	Precision	Recall	F1
Val	0.8731	0.8438	0.8201	0.8318
Test	0.8787	0.8479	0.8324	0.8401

Table 5: Random Forest Performance

Split	Accuracy	Precision	Recall	F1	Loss
Val	0.9053	0.9033	0.8428	0.8720	0.3570
Test	0.9084	0.9097	0.8443	0.8758	0.3418

Table 6: RoBERTa Model Performance

On the test set, performance is very similar: accuracy increases to 0.9084 and F1 to 0.8758, with precision 0.9097 and recall 0.8443. The loss values are consistent with these scores and indicate that the model’s predictions are generally confident in the correct label.

These small differences between validation and test metrics suggest that the fine-tuned RoBERTa model generalizes well to unseen data. Training loss decreases to around 0.13 over five epochs, while test loss stabilizes around 0.34, indicating that the model has learned a stable decision boundary without severe overfitting.

3.4 Training Dynamics

Figure 2 and Figure 3 show training loss and validation loss over the course of fine-tuning RoBERTa. Training loss decreases roughly linearly from about 0.29 to 0.13 over 9 steps (0 to 8). Validation loss starts near 0.27, rises to a peak of roughly 0.5 midway through training, and then drops to around 0.36 at the final evaluation. These plots summarize the optimization behavior of the model across epochs and the range of validation loss values observed during fine-tuning.

Figure 4 demonstrates the validation F1 score at each evaluation step. F1 fluctuates in a narrow range between approximately 0.8695 and 0.872 over the course of the model’s training. After a small dip at the intermediate evaluation step, F1 returns to and slightly exceeds its earlier values, reaching about 0.872 at the final evaluation.

On the test set, the final RoBERTa model correctly classifies 3,104 non-clickbait headlines and 1,713 clickbait headlines, with 170 non-clickbait headlines predicted as clickbait and 316 clickbait headlines predicted as non-clickbait (see Table 7). Therefore, we can safely conclude that most predictions lie on the diagonal of the confusion matrix, with comparatively fewer off-diagonal errors.



Figure 2: Train Loss

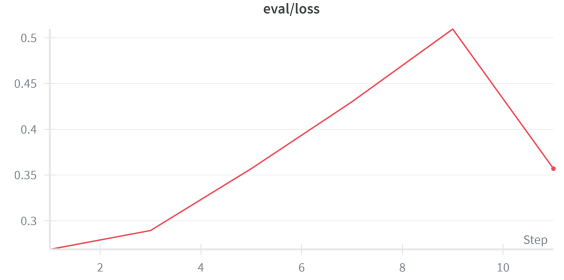


Figure 3: Eval Loss

These counts correspond to the precision, recall, and F1 values reported previously.

4 Discussion

4.1 Baseline Models

In order to properly evaluate the effectiveness of a fine-tuned model, it is important to establish a baseline for performance using more simple but effective ML models. This project establishes a baseline by comparing the performance of two separate models: logistic regression and random forest. In the results section, the models were evaluated on four key metrics used to determine the model’s capability of discriminating between clickbait and non-clickbait titles: accuracy, precision, recall, and F1.

4.1.1 Logistic Regression

Accuracy is a measure of the model’s ability to determine the true label of a sample over all of its predictions. In terms of clickbait, that would be the number of times that a model labels a “clickbait” article as “clickbait” and the number of times it models a “non-clickbait” article as “non-clickbait” over the total amount of predictions. The logistic regression model maintained an accuracy level of 0.8870 in the validation period and 0.8903 in the evaluation period. The model was able to correctly identify the true label of samples 88.7% and

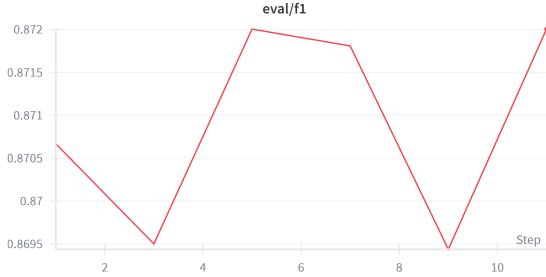


Figure 4: Eval F1

	Pred: False	Pred: True
Actual: False	3104	170
Actual: True	316	1713

Table 7: Confusion Matrix for RoBERTa on Test Set

89.03% of the time, respectively. The similarity between the values suggests that the model remains stable over unseen data.

In the scope of this paper, precision is a measure of how often the model was able to correctly identify a “clickbait” article as “clickbait” over the total number of times it labeled a sample as “clickbait.” The logistic regression model maintained a precision level of 0.8418 in the validation period and 0.8473 in the evaluation period. The closeness of these numbers suggests that the model was able to maintain its precision capabilities even over new samples. Compared to the accuracy measure, the logistic regression model tended to overestimate the amount of “clickbait” samples in the data.

Recall refers to the amount of “clickbait” titles labeled “clickbait” compared to the actual amount of “clickbait” titles. The logistic regression model managed a score of 0.8679 in the validation period and 0.8699 in the evaluation period. Seeing as these numbers are different by about 0.002, it is safe to say that the logistic regression model did not dip in performance when run on the evaluation set. Generally, the logistic regression model did a better-than-average job at identifying “clickbait” samples.

The F1 model is used to provide a balanced score between precision and recall. In this paper, the fine-tuned model must maximize this score, as the dataset that is used has around 50% more negative samples (“non-clickbait”) compared to positive samples (“clickbait”). By maximizing this score, we can verify that the model performs well on both kinds of samples, instead of showing a bias for one

over the other. When scored on F1, the logistic regression model performs well with 0.8546 on the validation set and 0.8585 on the evaluation set. These scores are very similar, suggesting that the model did not deviate in performance when tested on the evaluative set. The numbers also suggest that the model is relatively well-balanced without much favoritism.

Overall, the logistic regression model scores well on each of the components used to measure performance, setting a high bar for the fine-tuned model. Even when trained on just TF-IDF features, models (like logistic regression) do a good job at correctly classifying “clickbait” and “non-clickbait” titles.

4.1.2 Random Forest

As stated in the methods section, the random forest baseline was included because of its non-linear nature, which utilizes decision trees to make calculations that linear models may miss. The addition of this model greatly increases the solidity of the baseline, which also amplifies the significance of scoring above the baseline.

When evaluated on accuracy, the random forest model scored 0.8731 in the validation period and 0.8787 in the testing period. This is slightly less than the performance measured in the logistic regression model, which scored more than a full percent higher than the random forest model in both periods. These scores are still pretty high, despite being slightly worse than the logistic regression model.

In terms of precision, the random forest model measured at 0.8438 in the validation period and 0.8479 in the testing period. The performance is slightly better in the testing period, suggesting that the model does not dip in performance when introduced to new samples. In comparison to the logistic regression model, the random forest model performed slightly better at precision, meaning that its ratio of true positives to all positives was slightly higher than that of the logistic regression model.

When measured on recall, the random forest model scored 0.8201 on the validation set and 0.8324 on the test set. In relation to the other measures of performance, the random forest model performs worse on recall than on both accuracy and precision, suggesting that it struggles in this area. This is also true when compared to the logistic regression model, which is able to score a few percentage points higher in both sets compared to the random forest model. These results show that

recall is not this model's strong suit.

The F1 score is the most important measure of performance in relation to our models and dataset. For the random forest model, it managed a score of 0.8318 in the validation period and 0.8401 in the testing period. While this is generally a good score, it is still worse compared to the logistic regression model, which is able to score slightly higher in both periods.

While seen as the more complex model, the random forest model consistently performs worse on the measures of performance when compared to the logistic regression model, which is void of complicated components like decision trees. Despite this, its deviation from the scores set by the logistic regression model is able to provide a more nuanced baseline for the fine-tuned models to be compared against, strengthening the validity of their performances.

4.2 RoBERTa Model

Of the three models that were fine-tuned for the clickbait classification task, it was found that RoBERTa had the best F1 score and overall results compared to the other models. When measured against the baseline models, the RoBERTa model outperforms in nearly every category, with the exception of recall against the logistic regression model. The starkest improvement is in regard to precision, which saw a bump of 6% from the scores of the baseline models. The F1 score, a fairer examination of the model's capabilities, also showed improvement, scoring 0.8720 on the validation set and 0.8758 on the test set. The slight increase in score allows us to conclude that the model is capable of maintaining performance against unseen samples. The model scored better in this category than all of the baseline models and the other fine-tuned models (BERT and DistilBERT), meaning that its classification habits were the least biased amongst all tested models. The reasoning for this could point to a combination of the fine-tuning process and the mechanics that RoBERTa is built on.

The RoBERTa model also returned its loss, or the measure of how far off the model's prediction of the label was from the ground truth. The model's loss was slightly better in the evaluation period (0.3418) compared to the validation period (0.3570), pointing to the model's ability to maintain its performance in regard to this metric. While the loss steadily declines throughout the validation period,

there is a spike upwards in the evaluation period. This is normal behavior, as the test set includes unseen data, which can be harder for the model to predict.

Chakraborty et al. (2016) trained a model on clickbait classification and was able to generate an accuracy of around 93%. In comparison, the RoBERTa model achieves a clickbait detection accuracy of 90.53% on the validation set and 90.84% on the test set. While RoBERTa is able to outperform the baseline models and other models that have been fine-tuned on the same dataset, it is still outperformed by a separate model trained for the same purpose. There are several possible reasons for this disparity. One such is that the difference in outcome can be attributed to the difference in method. The model used in Chakraborty et al. (2016) was trained using various Naive Bayes classifiers and support vector machines, while the fine-tuning method used on the RoBERTa model relied on a unified sweep configuration to identify effective hyperparameters. It is possible that the Naive Bayes approach is better for this specific task, giving that model a slight edge over the fine-tuned RoBERTa model. Another possible reason for the gap is the difference in training data. While the datasets used to train both models are relatively the same size, the dataset used to train the RoBERTa model had an imbalance of samples, with around 50% more "non-clickbait" samples than "clickbait" samples. It is possible that the dataset used to train the model in Chakraborty et al. (2016) had a more balanced set of samples, allowing it to better understand the differences between the two classes.

5 Conclusion

5.1 Key Insights

In the age of the internet, users are subjected to biased headlines written for maximum engagement rather than objective information. This threatens media literacy, as users may engage with online articles that are misleading or lacking in factual data. The purpose of fine-tuning a pre-trained model on clickbait classification was to generate a tool that could guide users away from misinformation and improve their online habits. Labels that warn consumers of potential wrongdoing in articles may dissuade them from consuming that type of media, leading to a better comprehension of current events. The results of the performance of the RoBERTa model on the evaluation data show that the model

is generally able to accurately identify “clickbait” headlines in an unbiased manner. This model was built using data and mechanisms that were readily available online, pointing to the capabilities of others to build similar models. The fine-tuned RoBERTa model can be used by others to better regulate their consumption of online media.

5.2 Limitations

Despite our model’s strong performance, our approach has several limitations. First, the News Clickbait Dataset consists only of English news headlines collected from a particular set of outlets, so we likely won’t be able to generalize its results to other domains such as social media posts, entertainment content, or non-English data. The dataset is also moderately imbalanced, with substantially more non-clickbait than clickbait examples, which may contribute to RoBERTa’s comparatively lower recall relative to precision. In addition, our hyperparameter sweeps were run on a 2,000-sample subset of training data and for a limited number of epochs, meaning we may not have explored the full space of potentially better configurations. Finally, we treat clickbait detection purely as a binary sentence-level task over headlines, which keeps the problem simple but ignores several layers of complexity. For example, we do not study edge-case headlines with tricky phrasing or toned-down clickbait markers, nor do we analyze fairness across different topics such as entertainment, health, and politics, where clickbait rates may differ. We also do not incorporate article bodies, teaser text, images, or user-engagement signals, and we rely on a single yes/no label rather than a more fine-grained scheme that distinguishes different degrees or types of clickbait.

5.3 Future Work and Connection to Prior Research

Taken together, our results position this project between two strands of prior work on clickbait detection. Earlier systems relied on classical machine-learning methods such as Naïve Bayes, SVMs, logistic regression, and random forests built over hand-engineered features or multi-field representations of articles. More recent work has explored prompting large language models without any task-specific fine-tuning. However, these prompt-only systems still underperform specialized transformers on clickbait classification. In contrast, our fine-tuned RoBERTa model offers a practical middle

ground, as it consistently outperforms established baselines, while remaining compact enough to be used as a spam-filtering component in real moderation pipelines.

At the same time, the limitations above highlight several natural directions for future work. One avenue is to test how a robust fine-tuned transformer remains under intentionally tricky rewrites of the same headline, such as introducing sarcasm, neutralizing, or polarizing punctuation. Another direction is to expand coverage beyond a single domain, and evaluate whether performance remains consistent across a wide range of social media mediums, and whether separate domain-specific models or a single shared model works better.

Beyond the headline-only setting, there is also room to reconnect with multi-field architectures from earlier clickbait research by including article bodies, teaser text, images, or user-interaction signals alongside the titles. Finally, because we care about helping readers navigate online information, it would be valuable to move from offline metrics to user-facing experiments, such as deploying the model in a browser extension or news platform and collecting feedback on which predictions users find helpful or misleading, then combining this with interpretability tools for transformers to better understand which linguistic cues the model relies on. These extensions would turn our current proof-of-concept into a more general and deployable LLM-based solution for mitigating the harms of clickbait.

References

- Angela Bazaco, Marta Redondo García, and Pilar Sánchez-García. 2019. [Clickbait as a strategy of viral journalism: conceptualisation and methods](#). *Revista Latina de Comunicacion Social*, 74:94–115.
- Abhijnan Chakraborty, Bhargavi Paranjape, Sourya Kakarla, and Niloy Ganguly. 2016. [Stop clickbait: Detecting and preventing clickbaits in online news media](#). In *2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pages 9–16.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [Bert: Pre-training of deep bidirectional transformers for language understanding](#). *Preprint*, arXiv:1810.04805.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. [Roberta: A robustly optimized bert pretraining approach](#). *Preprint*, arXiv:1907.11692.

Martin Potthast, Sebastian Köpsel, Benno Stein, and Matthias Hagen. 2016. [Clickbait detection](#). volume 9626, pages 810–817.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2020. [Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter](#). *Preprint*, arXiv:1910.01108.

Peter Suci. 2020. [From scams to mainstream headlines, clickbait is on the rise](#).

Han Wang, Yi Zhu, Ye Wang, Yun Li, Yunhao Yuan, and Jipeng Qiang. 2025. [Clickbait detection via large language models](#). *Preprint*, arXiv:2306.09597.

Appendix

GitHub Link

Our GitHub repo link is: <https://github.com/samuelllee77/clickbait-classification>

Contribution Statement

- **Samuel Lee:** Designed the experiment setups and wrote the repository code; helped format the tables, figures, and reference in the report with latex.
- **Michael Pena:** wrote the results sections of the report; helped conduct the initial exploratory data analysis and generate the visualizations of the results.
- **Anna Lee:** wrote the methods section of the report; helped train and test the baseline models of logistic regression and random forest.
- **Sravya Chittuluri:** wrote the abstract and introduction sections of the report; helped with hyperparameter tuning part.
- **Gillian Wentzel:** wrote the discussion and conclusion sections of the report; helped train the final model.